automate the chase, own the onboarding.

bilingual · automated · audited

① system

workflows
4

DB tables
6

trigger
schedule + manual

error net
Workflow 0

stack
n8n + PG 18

messages
bilingual code

② scale

hires
20

tasks
253

factory
6 × 14 = 84

field force
6 × 13 = 78

head office
7 × 13 = 91

blocked
1 (NULL mgr)

③ output

unapproved
0

languages
Bangla + EN

audit rows
4

error rows
2

blocked rows
2

data
synthetic

what it does – no.01	No.01
workflows	4 (0+1+2+3)
tables	6
hires loaded	20
tasks seeded	253
intake trigger	manual per hire
daily check	08:00 AM
approval poll	every 15 min
overdue threshold	3 days
escalation route	approval queue
routine route	direct → outbox
bilingual rule	preferred lang='bn'
error catcher	Workflow 0
audit events	blocked + errors
messages	code node (no API)
outbox gate	pass if 0 unapprvd
data	synthetic

1 missed deadlines

- task owners receive no automated reminders
- onboarding slips silently past due date
- HR has no visibility into status until review
- no escalation path exists for critical tasks

2 no escalation path

- blockers only surface at probation review
- no approval gate for overdue critical tasks
- overdue ≥ 3 days has no defined owner
- no way to distinguish urgent from routine

3 language barrier

- English-only comms exclude factory workers
- Bangla-preferred hires miss key instructions
- no bilingual messaging exists in current flow
- compliance risk: instruction not understood

4 no audit trail

- blocked hires leave no compliance record
- workflow errors are invisible to HR
- no log of messages sent or approvals give
- zero traceability for disputes or reviews

hire type context · No.02 No.02

head office	7 hires
factory	6 hires
field force	6 hires
blocked	1 HO hire
blocker rule	NULL mgr id
language bn	Bangla msg
language en	English msg
escalation	≥ 3 days OD
routine	< 3 days OD
approval	required
direct	routine only
error log	audit log
processed	BOOLEAN flag
total tasks	253
max/hire	14 (factory)
min/hire	13 (field/HO)

0 Error Logger

- Error Trigger
- catches W2 & W3 failures
- writes to audit_log

1 Intake

- manual trigger
- validate hire
- generate task plan
- detect blockers

2 Daily Check

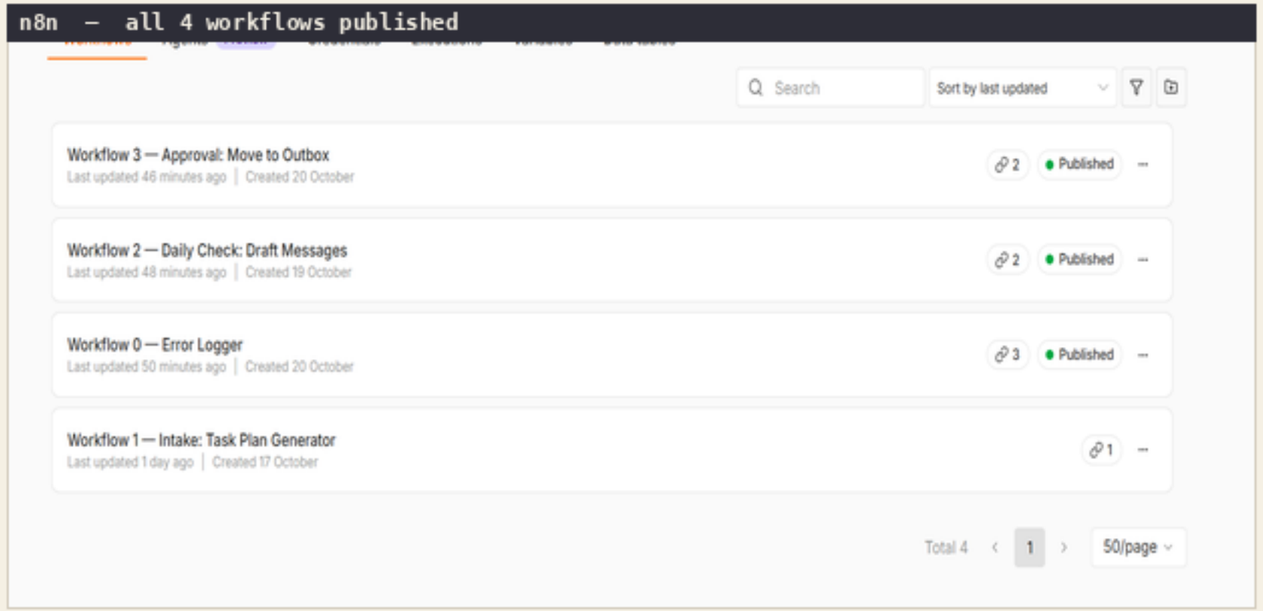
- 8:00 AM daily
- fetch overdue tasks
- route by severity
- bilingual

3 Approval

- every 15 min
- move approved escalations to outbox
- mark processed

data flow: hire → W1 intake → tasks → W2 daily → route → W3 approval → outbox
any W2/W3 error → W0 error logger → audit_log event_type='error'

n8n — all 4 workflows published



database schema — 6 tables

hires
hire records, type, language pref

task_templates
master task list per hire type

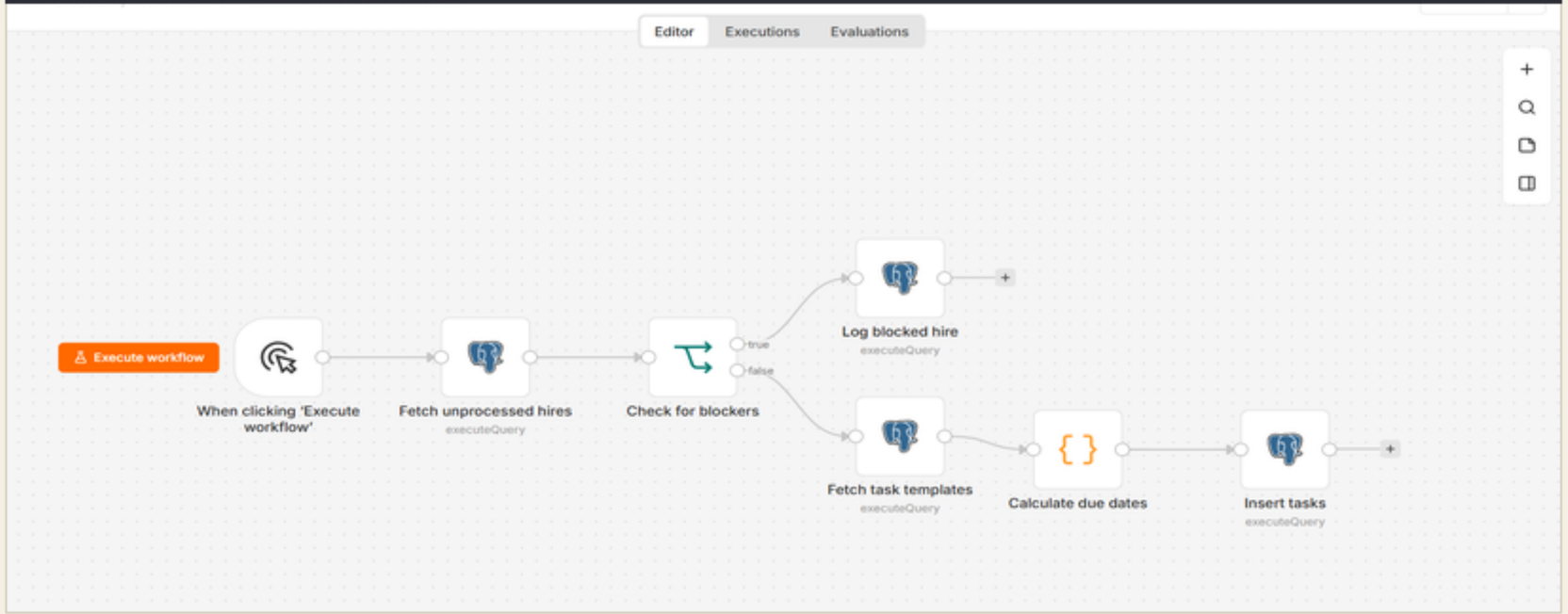
tasks
per-hire instances, due dates, status

approval_queue
escalations pending HR sign-off

outbox
messages cleared for sending

audit_log
blocked hires & workflow errors

Workflow 1 canvas — 6 nodes — manual trigger



nodes

① fetch hires
WHERE task plan not generated

② check blockers
IF head_office AND mgr_id IS NULL

→ **blocked**
log to audit_log · HALT

→ **clear**
continue to templates

③ fetch templates
SELECT WHERE hire_type matches

④ calc due dates
code node · offset by task_order

⑤ insert tasks
bulk INSERT into tasks

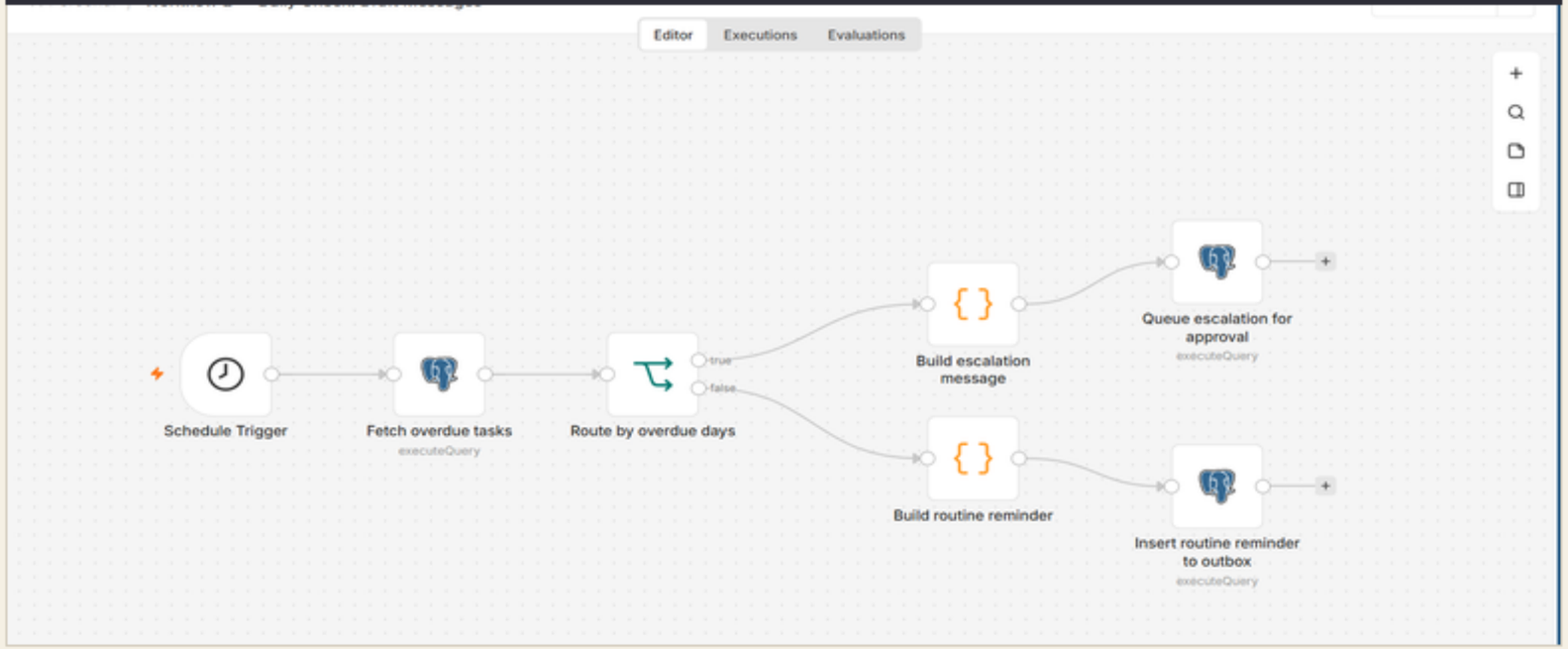
⑥ mark done
UPDATE hires.plan_generated=true

blocker rule: hire_type = 'head_office' AND manager_id IS NULL → INSERT audit_log · HALT · no tasks created

result: 1 hire blocked (hire_id=5) · remaining 19 hires processed · 253 tasks inserted

253	20	1	14	13	13
tasks seeded	hires processed	blocked hire	tasks/factory	tasks/field	tasks/HO

Workflow 2 canvas — 7 nodes — 08:00 AM schedule



assumptions · No.05	No.05
trigger	08:00 AM daily
fetch query	status=pending
overdue calc	NOW()::date − due date
threshold	3 days
branch A	overdue ≥ 3 → escalation
branch B	overdue < 3 → routine
escalation dest	approval queue
routine dest	outbox (direct)
bilingual rule	preferred lang='bn'
Bangla path	Bangla message body
English path	English message body
error handling	W0 catches any fail

routing logic

overdue ≥ 3 days:

→ build escalation message (bilingual)

→ INSERT into approval_queue

→ awaits HR sign-off before outbox

overdue < 3 days:

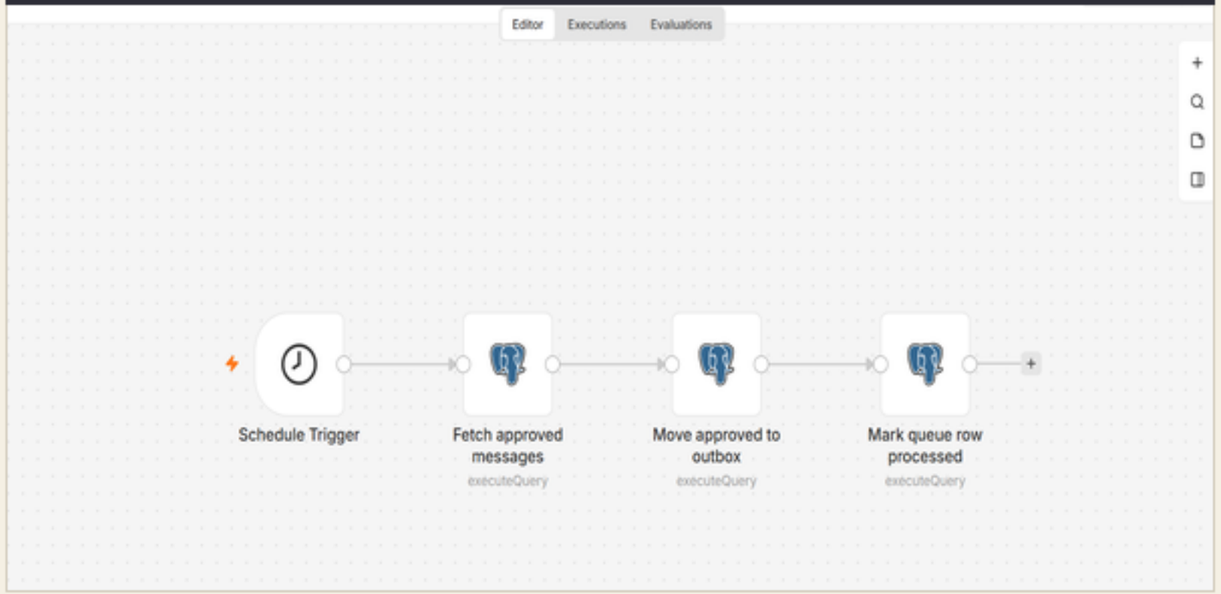
→ build routine reminder (bilingual)

→ direct INSERT into outbox

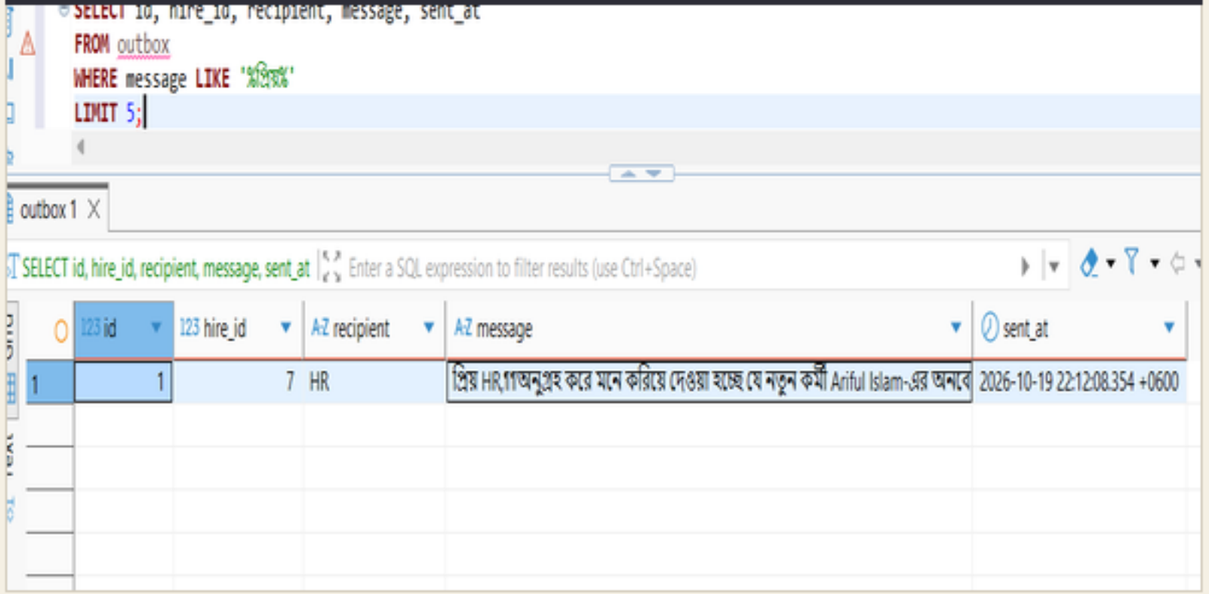
verdicts

bilingual rule	pass
no API dependency	pass
approval gate live	pass
W0 error net	pass
processed flag	pass

Workflow 3 canvas — 4 nodes



outbox — Bangla escalation message confirmed



1. fetch approved

```
SELECT from approval_queue

WHERE status = 'approved'

AND processed = false
```

2. move to outbox

```
INSERT into outbox

message, hire_id, sent_at

copy from approval_queue row
```

3. mark processed

```
UPDATE approval_queue

SET processed = true

idempotent — safe to re-run
```

verdicts

0 unapproved in outbox pass

Bangla msg confirmed pass

approval gate enforced pass

processed flag set pass

audit_log — 2 error rows (event_type='error') + 2 blocked rows

```
SELECT id, event_type, detail, logged_at
FROM audit_log
ORDER BY logged_at DESC;
```

audit_log 1

SELECT id, event_type, detail, logged_at FROM | Enter a SQL expression to filter results (use Ctrl+Space)

Grid | Text

	id	event_type	detail	logged_at
1	4	error	Example Workflow: Example Error Message	2026-10-20 14:31:31.780 +0600
2	3	error	Example Workflow: Example Error Message	2026-10-20 14:31:28.799 +0600
3	1	blocked	hire_type=head_office, manager_id=NULL	2026-10-19 12:37:58.220 +0600
4	2	blocked	hire_type=unknown_type, manager_id=MGR001	2026-10-19 12:37:58.220 +0600

assumptions

253 tasks seeded	factory 84 · field 78 · HO 91
20 hires loaded	7 HO · 6 factory · 6 field · 1 blocked
bilingual routing	hire pref lang='bn' → Bangla in outbox
error trigger live	W0 catches W2/W3 failures
blocked hire logged	HO + NULL mgr id → audit log row
outbox gate	0 unapproved allowed through
processed flag	idempotent — no duplicates

verdicts

0 unapproved in outbox	pass
bilingual confirmed	pass
blocked hire logged	pass
errors caught & logged	pass

253	20	4	6	2	0
tasks seeded	hires	workflows	DB tables	languages	unapproved